



EMPIRICAL RESEARCH

Developing information processing capability for operational agility: implications from a Chinese manufacturer

Pei-Ying Huang¹, Shan L. Pan²
and Tao Hua Ouyang³

¹Department of Information Systems, School of Computing, National University of Singapore, Singapore; ²Australian School of Business, University of New South Wales, Australia; ³School of Economics and Management, Beihang University, Beijing, China

Correspondence: Tao Hua Ouyang,
School of Economics and Management,
Beihang University, 37 Xueyuan Road,
Beijing, 100191, PRC.
Tel: +86-18601259357;
Fax: +86-10-82338631;
E-mail: taohuaouyang@hotmail.com

Abstract

The operational agility literature suffers from a lack of clarity in terms of the process or underlying mechanisms through which operational agility is achieved. Drawing on the information processing view of the firm, this study attempts to address this gap by examining the process of operational agility development. A process model of developing information processing capability for operational agility is proposed. As the model suggests, operational agility is achieved through a two-step process – the construction of IT-enabled information processing network and the implementation of organizational control – to enhance the right information processing capability. We further identify three types of information processing capability – information sensitivity, information synergy and information fluidity – that enable operational agility. Furthermore, a five-step practical guide for developing information processing capability for operational agility is provided for practitioners. This is achieved by conducting a case study of Haier, one of the largest household appliance manufacturers in China. This paper concludes with a discussion of the potential contributions and directions for future research.

European Journal of Information Systems (2014) 23(4), 462–480.

doi:10.1057/ejis.2014.4; published online 18 March 2014

Keywords: operational agility; operational decision making; information processing capability; case study

Introduction

Operational agility is a capability that enables organizations to sense changes in turbulent business environments, as well as conceive appropriate competitive actions to seize market opportunities (Sambamurthy *et al*, 2003). It has been described as an important determinant of firm success in today's rapidly changing business environments (Dove, 2001). As a result, this concept has attracted much attention from industry and academia. Practitioners in United Kingdom have initiated the Operational Agility Forum since 2009. Senior business and IT professionals gather and debate how organizations could sense and respond more efficiently and effectively to a new business environment. Moreover, one of the largest professional service firms, Ernst & Young, has published a report to share some practical lessons for developing operational agility (Dalne, 2011). Meanwhile, many studies have been conducted by academic researchers to examine operational agility, contributing valuable findings to this body of knowledge (e.g. Sambamurthy *et al*, 2003; Raschke & David, 2005; Raschke, 2007; Tallon, 2008; Wu & Li, 2008).

Despite the increased knowledge of operational agility achieved in the past few years, the fundamental question of how operational agility is

Received: 29 January 2012
Revised: 30 September 2012
2nd Revision: 3 May 2013
3rd Revision: 3 July 2013
4th Revision: 29 December 2013
Accepted: 5 February 2014

developed remains unanswered. Existing studies mainly center on identifying and validating the antecedents of operational agility. IT has generally been considered as an enabler of operational agility (Sambamurthy *et al*, 2003). Other relevant antecedents that have been studied include digital options (Sambamurthy *et al*, 2003), IT infrastructure (Raschke, 2007), and IT capabilities (Tallon, 2008; Wu & Li, 2008). These studies share a common assumption that treats the relationship between IT and operational agility as a 'black box', and implicitly presumes it to be automatic. However, the assumption might not be true, since the impact of IT is limited unless it is aligned with the business processes of an organization (Porter, 2001). Some studies even offer a rebuttal to this assumption. It is found that the relatively fixed physical and technological artifacts of information systems may actually hinder and even impede agility (Galliers, 2007). This is contradictory to the findings that endorse IT as an enabler of operational agility. The inconsistent findings in the literature challenge studies that treat the relationship between IT and operational agility as a 'black box' and reveal a lack of clarity of the process through which IT enables operational agility. Therefore, there is a need to investigate the process of operational agility development and open up the 'black box' that lies between IT and operational agility.

This study draws on the lens of information processing to examine the development of operational agility. Effective information processing enables firms to acquire relevant, accurate and comprehensive information in a timely and cost-economic manner (Mithas *et al*, 2011), leading to more efficient communication and rapid decision making, which is essential for operational agility (Sambamurthy *et al*, 2003; Glenn, 2009). The ability to process information is significantly increased through the application of information systems (Mani *et al*, 2010). Hence, the lens of information processing offers a sound means to observe the process through which IT enhances operational agility. The information processing view of the firm characterizes firms as information processing systems, faced with various types of task uncertainty (Galbraith, 1973; Tushman & Nadler, 1978). The objective of the firm is to develop the right information processing capability that facilitates the right amount of information needed to cope with uncertainty, and achieve desired performance (Mani *et al*, 2010). Accordingly, we theorize that the achievement of operational agility requires firms to develop appropriate information processing capabilities, to reduce uncertainty and thereby enhance sensing and responding capabilities.

The research question of this study is: *How to develop information processing capability for operational agility?* Informed by the information processing view of firms, the information processing network (e.g. Ahuja & Carley, 1999; Kwon *et al*, 2007) and organizational control (e.g. Leifer & Mills, 1996; Turner & Makhija, 2006) are examined as two strategies to enhance information processing capability. We adopt qualitative methods and conduct a case study of the largest household appliance manufacturer in China. The findings show that operational agility

is achieved through a two-step process – the construction of information processing network and the implementation of organizational control – to enhance the right information processing capability. Furthermore, we identify three types of information processing capability – information sensitivity, information synergy and information fluidity – that enable operational agility. On the basis of the theoretical findings, a five-step practical guide of developing information processing capability for operational agility is provided for practitioners.

Literature review

Operational agility

Operational agility refers to 'the ability of firms' business processes to accomplish speed, accuracy and cost economy in the exploitation of opportunities for innovation and competitive action' (Sambamurthy *et al*, 2003, p. 245). It highlights the ability of firms' internal business processes to rapidly cope with market or demand changes (Lu & Ramamurthy, 2011). Operational agility is considered as an important determinant of a firm's success in today's rapidly changing business environments (Dove, 2001). A prominent real-world example that capitalizes on operational agility for business success is that of Zara, one of the biggest clothing retailers in the world. Its success is founded on a super-responsive supply chain, which enables the company to deal with unpredictable and quick changes of customers' tastes in the highly volatile fashion garment industry. This 'fast fashion' system depends on effective information processing, which allows a constant exchange of information throughout every part of Zara's supply chain, and offers real-time information to support rapid decision making (Ferdows *et al*, 2004).

Past research on operational agility mainly centers on identifying and validating the antecedents of operational agility. Sambamurthy *et al* (2003) posit IT as an enabler of operational agility. They argue that IT is applied to establish electronic communication channels, which in turn enable a rapid and up-to-date supply of comprehensive information to enhance operational agility. Some empirical studies on operational agility apply variance models to investigate the impacts of antecedent constructs on operational agility, such as research investigating the effects of a firms' IT infrastructure on operational agility (Raschke, 2007), a study examining the relationship among environmental dynamism, managerial IT capabilities, technical IT capabilities and operational agility (Tallon, 2008), and the research exploring the mediating effects of operational agility on the relationship between IT capability and firm performance (Wu & Li, 2008). It seems that they generally treat the relationship between IT and operational agility as a 'black box', and implicitly assume it to be automatic.

However, there are some studies that offer a rebuttal of this assumption. These studies indicate that greater investments in process and IT may hinder, and even impede agility (Weill *et al*, 2002; Overby *et al*, 2006), partly due to the relatively fixed physical and technological artifacts of

information systems (Galliers, 2007). Inflexible IT systems, rigid IT architectures or disparate technology silos often result in constraints on businesses (van Oosterhout *et al*, 2006). As a result, IT may become a disabler for agility (Lu & Ramamurthy, 2011). This is contradictory to the findings that endorse IT as an enabler of operational agility. The inconsistent findings in the literature challenge studies that treat the relationship between IT and operational agility as a 'black box', revealing a lack of clarity in the process through which IT enables operational agility. Therefore, this study attempts to open up the 'black box' between IT and operational agility, and close the theoretical gap by investigating the process of operational agility development.

Information processing view

The information processing view of the firm posits that firms need quality information to cope with uncertainty and improve their decision making (Galbraith, 1973). This theory identifies three important concepts – information processing requirement, information processing capability and the fit between the two to obtain optimal performance (Galbraith, 1973). The information processing requirement is the amount of information about organizational activities that is needed to address uncertainty (Tushman & Nadler, 1978). Information processing capability refers to the ability to gather, synthesize and disseminate information properly to cope with uncertainty (Tushman & Nadler, 1978). Uncertainty stems from three sources, namely environmental dynamism, task interdependence and task complexity (Tushman & Nadler, 1978; Gattiker, 2007). High levels of uncertainty defer decision making (Wang *et al*, 2013), impairing the efficiency and effectiveness of sensing and responding activities.

Typically, firms have two strategies to cope with uncertainty and increased information needs. First, firms can increase adaptability by implementing advanced information processing mechanisms to reduce uncertainty (Galbraith, 1974). Information processing mechanisms that provide real-time, accurate and relevant information allow firms to be flexible in adapting activities at the time of uncertain event (Mithas *et al*, 2011). An example of this strategy is the implementation of integrated information systems that enhance information flow and reduce uncertainty (Premkumar *et al*, 2003). Second, firms can enhance the predictability of task execution by pre-specifying agreed-upon actions to reduce the effect of uncertainty (Galbraith, 1974). The approach to pre-specify actions, formal or informal, can guide and motivate organizational members to achieve congruent judgment in the face of uncertainty (Ouchi, 1979), and thereby facilitate decision making. A classic example of this strategy is the development of rules or standard operating procedures to reduce the effect of uncertainty (Tushman & Nadler, 1978). The above two strategies enable rapid response to environmental change (Wang *et al*, 2013), and thus they are significant for the development of operational agility. Both of them are reflected in the constructs described below.

Information processing network is defined as a dynamic network-based information processing structure, which operates as a coordination mechanism that transcends formal hierarchy (Kwon *et al*, 2007). It often assumes the role of backbones that support information and knowledge-based activities within and across firm boundaries (Owen-Smith & Powell, 2004). Since effective information gathering, synthesizing and disseminating of an information processing network is essential for survival and competence (Kodama, 2005), firms should construct and refine information processing networks to maintain an optimal information flow (Kwon *et al*, 2007). With the step-shift advances in IT over the past few decades, the potential of IT to create and maintain flexible information processing networks has been demonstrated by researchers and practitioners (e.g. Jarvenpaa *et al*, 1998; Fulk & DeSanctis, 1999; Kraut *et al*, 1999; Wiesenfeld *et al*, 1999; Kwon *et al*, 2007; Pan *et al*, 2012).

The structure of networks has been used by prior research as a main dimension to study information processing networks (e.g. Kwon *et al*, 2007; Pan *et al*, 2012). The structure of an information processing network depicts the patterns through which organizational communication is expedited and information is processed (Ahuja & Carley, 1999). Existing studies have examined the level of hierarchy and the degree of centralization of information processing networks (e.g. Ahuja & Carley, 1999; Kwon *et al*, 2007). Hierarchical levels are reflected by the number of levels one must go through in order to obtain information (Hummon & Fararo, 1995). Centralization indicates the extent to which a network or group is organized around its focal point (Freeman, 1979). While an information processing network with high levels of hierarchy and centralization is well suited to tasks that are routine and with low levels of interdependence, the information processing network with low levels of hierarchy and centralization is more efficient in dealing with tasks that demand flexibility and adaptability (Baker, 1992; Ahuja & Carley, 1999).

Organizational control is defined as all attempts to ensure that individuals behave in a manner consistent with organizational objectives (Ouchi, 1979; Ouchi, 1980). Organizational control encompasses routines, coordination mechanisms or organizational norms (Turner & Makhija, 2006). It aligns individual capabilities, activities and performance with the organization's goals and aspirations (Cardinal *et al*, 2004). Organizational control facilitates information flow by influencing individuals' information processing behaviors (Leifer & Mills, 1996). It requires 'specific relationships between individuals and groups that influence how information is shared and knowledge is disseminated within the firm' (Turner & Makhija, 2006, p. 198).

Organizational control can be classified into four control modes – behavior control, outcome control, clan control and self control (Kirsch, 1997). Behavior control is exercised when proper behaviors are pre-specified and rewards are based on how well the behaviors are followed (Rustagi

et al, 2008). Outcome control specified desired goals without defining proper behaviors and rewards are based on whether the goals are achieved (Kirsch *et al*, 2002). As the appropriate behaviors and expected outcomes are pre-specified, behavior and outcome controls are associated with less flexibility necessary to cope with uncertainty (Leifer & Mills, 1996). Clan control operates when all members of the work group embrace a common set of norms and values. Rewards and sanctions are based on whether individuals behave in accordance with these norms and values (Kirsch *et al*, 2010). In self control, individuals define their own goals and processes for the task, and then proceed to self-monitor, self-reward and self-sanction (Kirsch *et al*, 2002). As clan and self controls use unwritten, unofficial values, norms and beliefs to guide individual behaviors, they provide more job discretion for individuals, and are characterized by higher levels of flexibility and responsiveness (Leifer & Mills, 1996).

To sum up, we propose that the achievement of operational agility requires firms to develop the right information processing capability to reduce uncertainty and thereby enhance the sensing and responding capabilities. Figure 1 shows a conceptual model of how operational agility is achieved. The construction of information processing network and the implementation of organizational control are two strategies for developing information processing capability. This model is applied to guide subsequent data collection and analysis.

Research methodology

To investigate the research question regarding how operational agility is achieved, we adopted a case research approach. There are two reasons for choosing the case study method. First of all, the research question is a 'how' question, appropriate to explore through case studies (Walsham, 1995). Second, both organizational information processing and operational agility are complex and multi-faceted phenomena that are embedded within an organizational context (Pentland, 1999), which makes it more suitable to examine the phenomena through relevant stakeholders' interpretations (Klein & Myers, 1999) rather than a quantitative approach.

We developed three criteria for case selection. First, the case company should be situated in a highly competitive business environment, so that there is a need to develop operational agility. Second, the company should have demonstrated its capability in leveraging business processes to detect and respond to market changes (i.e. operational agility), so that managerial knowledge regarding how to achieve operational agility can be extracted from the case. Third, the internal operations of the selected company should be complicated enough, so that it is necessary to develop appropriate information processing capability to deal with the associated uncertainty. This informs us that a large company is preferred. Accordingly, we chose Haier, the largest household appliance manufacturer in China, as our case company. The

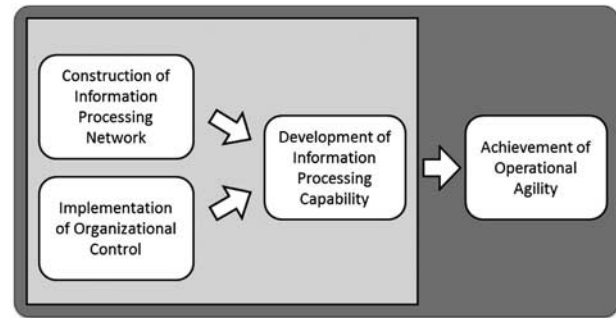


Figure 1 A conceptual model of how operational agility is achieved.

company is well-recognized for its responsiveness to market changes in the turbulent Chinese household appliance market. Arguably this might be due to its excellent order fulfillment process, which provides the company with flexibility and efficiency in adjusting its product designs and productions. Thus, this company is particularly appropriate for this study.

Data collection

The investigation took place over a 23-month period between May 2010 and March 2012. We followed the theory-building process as prescribed by Pan and Tan (2011) to design and conduct this study. The steps in our research are shown in Figure 2 (adapted from Pan and Tan (2011)). One of our authors has been tracing the growth of the case company for 6 years and collected a large amount of qualitative data as part of other studies. Therefore several discussions were conducted to make each author familiar with the case company and identify a set of themes that were potentially relevant to the research question. An archival analysis that took around 5 months complemented the discussions. A large amount of archival data from online documents, websites and books (e.g. Yong, 2008) were collected and analyzed. A mix of first and secondary data would increase the reliability of the study (Yin, 2003). Meanwhile we reviewed the relevant literature to predict a certain relationship between the theoretical lens and our phenomenon. These initial data collection and analysis allowed us to better understand the issues and, in combination with our review of the literature, confirmed the idea of applying the information processing perspective as an investigative lens for the phenomenon of interest. Accordingly an initial set of pertinent themes (such as information processing networks and organizational controls) was pre-identified. This resulted in the formation of our preliminary model (see Figure 1), which served as a 'sensitizing device' (Klein & Myers, 1999) to guide the following data collection and analysis.

Onsite data collection was conducted twice in September 2010 and November 2011. A series of group interviews with an average of three informants per group were conducted with the middle and top management (see Appendix A and Appendix B). According to Fontana and

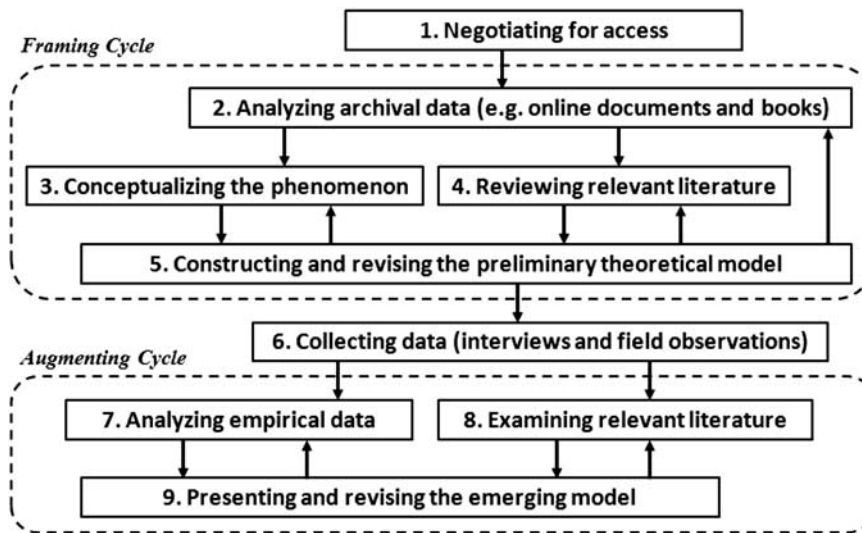


Figure 2 The steps of investigation.

Frey (2000), a group interview can produce ‘rich data that are cumulative and elaborative; they can be stimulating for respondents’ (p. 652). Thus, the type and range of qualitative data generated from group interviews are often deeper and richer than those collected from one-to-one interviews, which helps achieve a more comprehensive view toward the complex phenomena we attempt to study. The selection of participants for each group was based on the criteria that either they came from the same department, or they had attended the same project. Interviews were based on topic guides (see Appendix A), which demonstrated relevant probes at appropriate junctures (Pan *et al*, 2007). We adjusted the interview questions so that they were based on the organizational role of informants. Each informant was assured of the confidentiality of the data provided, especially when potentially sensitive information was sought (Walsham, 2006). The 10 group interviews, taking an average of 90 min were recorded digitally and later transcribed by one of the authors to form an equivalent single document of more than 100 pages (font size 10 pt and single line spacing). Given that the interviews were conducted in Chinese, the authors spent much time in translating the data. The group interviews were complemented by two individual interviews with the informant who was the most familiar with the process we were studying. The individual interviews also served to confirm our preliminary understanding, formed during group interviews. A follow-up interview was conducted with the director of Haier University and the director of Supply Chain Department in November 2011. The director of Haier University is responsible for staff training and he is very familiar with strategic and operational issues in the company. Therefore he was involved in the interview. The informants helped us verify and consolidate our findings. Site visits to the company’s facilities in the headquarters were arranged, and around 60 photos were

collected during site visits. Whereas the interviews formed the primary source of data, they were corroborated by other secondary data such as internal publications, organizational documents and field notes. Multiple data collection allows triangulation offering stronger substantiation of constructs and hypotheses (Eisenhardt, 1989).

Data analysis

Three coding techniques (i.e. open coding, axial coding and selective coding), which are commonly used in a qualitative study (e.g. Kirsch, 1997; Pan *et al*, 2006; Lim *et al*, 2012), were applied to analyze the interview data. Initially, the process of developing the information processing capability for operational agility was coded using open coding technique (Strauss & Corbin, 1990). Open coding starts by placing conceptual labels on the data to capture organizational activities that help explain the process. It centers on uncovering general recurrent themes. Some of the conceptual labels used in this study – such as *regular meetings* and *providing clear rules and procedures* – are control mechanisms suggested by the literature; others – such as *implementing strict schedule control* – were suggested by the data (Strauss & Corbin, 1990). In total, we developed 47 conceptual labels that matched our research objectives. The second step of data analysis was to link similar concepts from open coding into theoretical constructs, through axial coding (Strauss & Corbin, 1990). Associations among concepts were made primarily by referencing constructs from our preliminary model and the existing literature. Continuing with the previous example, *regular meetings* and *providing clear rules and procedures* are suggested by the literature as mechanisms to implement behavior control. The concept of *implementing strict schedule control* has not been discussed by the literature. It is executed by specifying a proper schedule to

follow. On the basis of this characteristic, we treated it as a mechanism to implement behavior control. As a result, we theorized the common underlying theme of these labels as the *implementation of behavior control*. Through axial coding, a meaningful comprehension of the phenomenon of interest was generated (Lim *et al*, 2012). Since the context of this study is the order fulfillment process, including order forecasting, sales and production planning, and goods production tasks, we differentiated the three tasks during data analysis to achieve three distinct sets of constructs. We provide a summary of axial codes in Appendix C. An example of the statistics of codes for group-6 interview is shown in Appendix D. Again, in the previous example, the hit of *implementation of behavior control* in the planning task is 58.3%, which indicates that more than half of the interviews have discussed this theme. As a last step, we used selective coding technique to integrate and refine concepts in order to ascertain a coherent picture of the phenomenon observed (Strauss & Corbin, 1990). In this step, the process of developing information processing capabilities for operational agility was selected as the core phenomenon. Furthermore, the constructs identified from axial coding were integrated and related to the core phenomenon. We accomplished this by examining back and forth the interview data, relevant literature and the emerging model. The emerging model was presented to several senior case researchers to generate feedback that helped us revise the model. This augmenting cycle (Pan & Tan, 2011) (steps 7, 8 and 9 of our investigation) lasted several months until theoretical saturation was reached, in which the findings of case study were comprehensively explained and no additional data could be collected or added to improve the emergent model (Eisenhardt, 1989).

Case description

Organizational background

Haier is the largest household appliance manufacturer in China, with over 240 subsidiary companies, more than 110 design centers, plants and trading companies throughout the globe. The company receives more than 0.9 million sales orders monthly. These orders involve more than 10,000 types of customized products, which require the procuring of more than 260,000 types of material. As the company grew rapidly, top management observed several problems of the order fulfillment process. Sales orders were predicted by salespersons based on prior experience. This information went through several levels of hierarchies to approach sales department in the headquarters, through which information was adjusted several times by different hierarchies. As a result, it was less likely for sales orders to be accurate. After this, the information was transferred to other departments (e.g. production, procurement and logistics departments) for fulfilling the orders. However, the cooperation across departments was inefficient, since information traveled across these departments one after another. This means that one department

has to wait for the input of another department to proceed. Therefore substantial time was wasted in the planning process. Production was based on inaccurate sales orders, which caused the problem of increasing inventory. This is a serious problem that might cause negative cash flow. Furthermore, as product category increased dramatically, the manufacturing process became increasingly complex. For example, it was difficult to manage raw materials, since some of them were general components used by different products, while some of them were specific to one product. As a result, the company started a large scale of business process re-engineering in 2008. This allowed the company to reshape its order fulfillment process. The overall operation improved significantly. The cycle of the order fulfillment process was shortened to 2 weeks, which is considered as best in the industry in China.

We are interested in the order fulfillment process changes. Specifically, we would like to examine how the company developed information processing capabilities to achieve operational agility throughout the order fulfillment process. Order fulfillment is a key process in managing the supply chain (Croxtan, 2003). This process can be defined as a complete process from the point of sales inquiry to the delivery of a product to customer (Fogarty *et al*, 1991). The tasks of order forecasting, sales and production planning, and goods production are three significant tasks in the order fulfillment process (e.g. Fliedner, 2003). Under the background of order fulfillment, we examine information processing practices that support the three sequential tasks. Data collected are presented according to the sequence of these tasks in the subsections that follow.

Forecasting task

The order fulfillment cycle in Haier begins with the order forecasting task. Sales department and salespersons try to predict customer demands over the next few weeks and generate sales orders for production. However, customer demands change unpredictably and quickly, which results in uncertainty of the task. The CEO of Haier described every customer as a 'flying target' (Yong, 2008, p. 27) to reflect the dynamic nature of customer demand changes. Furthermore, the need to handle a mass of customer demand information generated from more than 10,000 points of sales brings about increased uncertainty. As a result, the uncertainty of the order forecasting task requires the company to become sensitive to changes, so that it can capture accurate customer needs.

To become more sensitive to market changes, the company developed an information processing network that allowed efficient information collection from the local market. An example of this structure is the order management system. In the past, customer demands gathered by salespersons went through several levels of hierarchy to reach the sales department in the headquarters, through which information became distorted. This phenomenon could be vividly described as the 'bullwhip' effect. The

order management system was developed to solve this problem. It enables direct information exchange between salespersons that are close to local market and sales department in the headquarters. Hence, information distortion is largely eliminated. The headquarters also started to provide daily feedback information to salespersons via Short Messaging Service (SMS). This information processing network facilitates the centralization of market information from points of sales. Meanwhile it helps maintain a flat communication structure between salespersons and the headquarters.

The ability to capture customer changing needs is further enhanced by the implementation of a performance management program and employee empowerment. On the basis of information from the order management system, the company designed a performance management program called 'individual goal combination'. The program motivates salespersons to make more accurate decisions regarding customer orders that they are going to fulfill. It evaluates a salesperson's performance and classifies it based on the extent to which it is consistent with organizational goals. While desired performance is rewarded, undesirable outcome is punished in monetary terms. This program enhances a salesperson's responsibility to ensure the accuracy of decisions made. Furthermore, salespersons were given sufficient discretion in decision making, which guaranteed flexibility in combating demand variability. This is reflected in the belief that 'every salesperson is a Strategic Business Unit (SBU)' propagated by the company. Providing tools to support decision making helps to improve decision accuracy. An order forecasting system that integrated complicated forecasting models and a large amount of sales data was developed and used to assist in decision making.

Through the above practices, salespersons became more responsible. In order to establish a positive performance record, salespersons actively reach customers to understand their sales and inventory. They ask the sales department to offer analytical tools for order forecasting, so that they are capable of making more accurate sales decisions. This allows the generation of accurate market information, which is delivered to the headquarters in an efficient manner. As a result, the company became more capable in sensing changes of customer needs. The relevant themes identified under 'Forecasting Task' are summarized in Table 1.

Planning task

The sales and production planning task generates a congruent plan to coordinate the actions of diverse departments. Since the actions of different departments might depend on and influence each other, the task requires a tight cooperation between them. For example, the sales orders decided by the sales department influence the procurement plan of procurement department and the production plan of production department, so that they should communicate and adjust their plans accordingly. The interdependent nature of the planning task results in

the uncertainty that the company needs to cope with. Furthermore, different departments have diverse requirements and goals, which could be potentially conflicting, leading to increased uncertainty. Therefore, the company needs to facilitate effective communication and collaboration across departments to reduce the uncertainty of the planning task.

To achieve effective collaboration, the company constructed an information processing network that enabled efficient information exchange and integration across departments. The company restructured its IT infrastructures and integrated existing systems to boost extensive information exchange. For example, the company integrated around 600 existing systems, and established a unified data sharing platform to expedite information sharing. Furthermore, they developed an order review system to support the integration of relevant information from different departments. The order review system is used to check the alignment between requirements and constraints of different departments so that decisions made based on it will be consistent across departments. The system saves them much time and effort in resolving the redundancy and inconsistency of diverse departments' activities. This information processing network is decentralized as it facilitates information flow across departments. It is also a flat communication structure that allows mutual adjustment across departments.

The company implemented a series of clear procedures for coordination, and nurtured a collective culture to further enhance communication and collaboration across departments. A series of clear procedures guarantee sufficient communication and smooth potential conflicts across departments. These procedures formulated concrete time and activities to be performed by specific department as well as individual. For instance, the company conducted routine cross-departments meeting every Wednesday to communicate and negotiate sales and production issues. Except for the formal mechanisms, Haier's collective culture plays a key role in supporting the task. The company propagates common organizational value across departments. It also emphasizes individual obedience to 'one common goal' of the company. It promotes the belief that individual will benefit more when the goal of the company is accomplished. This culture is reflected by many slogans that the authors have observed in the case site. The collective culture facilitates collaboration by resolving conflicts across departments, which cannot be solved by pre-specified rules.

As a result, the company became capable in integrating information for effective communication and collaboration across departments. This facilitates better decision making in sales and production planning task. As information is integrated across departments, so the coordination of allocated resources, activities and roles across departments will be improved, leading to cost economic coordination. The relevant themes identified under 'Planning Task' are summarized in Table 2.

Table 1 The relevant themes and data identified under 'forecasting task'

<i>Axial codes</i>	<i>Exemplary quote and open codes</i>
Construct information processing network	Director of Department 1: We provided a portal for product representatives (salespersons). They not only make sales orders, but also view sales information of their own customers They can see the amount of income and loss every day. They also can see new tasks assigned by the headquarters. If they have any problems, such as some problems in customer's model machine, they can feed back to the headquarters directly. (Open code: establish multiple communication channels between salespersons and the headquarters.) Director of Strategy Department: The information of the performance evaluation result will be sent via SMS to every salesperson daily. By doing this, even though they are on a business trip, they still can know their performance, so that they can adjust accordingly. (Open code: establish multiple communication channels between salespersons and the headquarters.)
Implement outcome control	Director of Department 2: If you can fulfill your plan completely, you can get all your salary and season rewards. If you want to get your salary, you have to make a sound sales plan, to sell profitable products and to sell fast. (Open code: implement comprehensive performance evaluation and provide incentives based on performance.) Director of Department 4: We realized that the most important thing we should do is to clarify the responsibility of salespersons. We need to record sales information of salespersons formally. We allow them to view their performance. How many orders are placed? How many products are sold? All the information is recorded. They are responsible for their sales. (Open code: define individual responsibility.)
Implement self control	Director of Strategy Department: Except for selling fast, selling more, and selling profitable products, we still have one principle, which is to be independent in managing the business and responsible for one's own profits and losses. (Open code: instill a sense of responsibility.) Project Manager of Department 4: The performance of a company is determined by the performance of its employees. If each employee is performing well, the performance of the company as a whole will be good. So we need to combine individual interests with the organizational objectives. (Open code: fusion of individual goals and organizational objectives.)
The role of information processing capability	CIO of Haier: By implementing these mechanisms (to manage salespersons), our salespersons are just like engines with extremely strong power, because these mechanisms are very motivating. (Open code: achievement of high staff morale.) Director of Haier University: The sales order is generated based on sales information, including the sales of previous week, the current inventory, and the cycle of inventory turnover. Therefore this information becomes more precise. (Open code: achievement of accurate prediction.)

Production task

As a household appliance giant with a broad product range, the company produces over 10,000 types of customized products monthly. The process of fulfilling large production orders involves many complex issues. For example, the company needs to decide on the distribution of production tasks across manufacturing plants efficiently, and these decisions are made based on the considerations regarding the production capacity of plants, shipping address of sales orders, availability of raw materials, and other relevant issues. The complex nature of the task becomes a source of uncertainty. Furthermore, the compressed production time leads to increased uncertainty. Thus, the ability to facilitate the efficient flow of information between headquarters and manufacturing plants is needed to reduce the uncertainty of the production task.

The company developed an information processing network that enabled rapid information dissemination of the

headquarters to facilitate cooperation between the headquarters and plants. Given the large scale of information needed to process, it is infeasible to handle it manually. The company's enterprise resource planning (ERP) system, which is named 'Global Value System (GVS)', plays a pivotal role in processing and disseminating information. It records information relating to raw material, product and production process. Through analyzing the capacity of each manufacturing plant, sales order details, inventory information and other relevant information, many decisions are made by the system automatically. The system further arranges a production plan for each plant, based on the information relating to that particular plant. Thus, decision making is conducted automatically, and manual interventions are largely eliminated. This information processing network allows centralized control of production information in the headquarters. Production information is distributed from the headquarters to each level of manufacturing plants to coordinate production task.

Table 2 The relevant themes and data identified under 'planning task'

Axial codes	Exemplary quote and open codes
Construct information processing network	Project Manager of Department 4: When the company becomes larger, responding slowly (to customer demand) is undesirable. To be fast, we have to use information technology to achieve process synchronization ... All the departments, including salespersons, production staff and research staff will receive orders at the same time. (Open code: implement internal systems integration.) Director of Infrastructure Operation and Maintenance Department: To facilitate the cooperation among departments, we made several changes to the existing information systems technically, such as establishing a bus sharing platform and the release of GVS. (Open code: create an information sharing platform.)
Implement behavior control	Director of Department 2: For the process between forecasting and production, we use the '161 week order' mechanism to adjust the frequency of the process and improve its efficiency ... After using this mechanism, the cycle of order fulfillment is reduced from 40 days to 17 days. Actually we control the frequency of our business. (Open code: implement strict schedule control.) Director of Department 1: We have a set of procedures and mechanisms to cope with coordination between sales and operation. Salespersons will report results of order prediction on every Monday; the headquarters will adjust the results of order prediction on every Tuesday; then estimate short-term order prediction on every Wednesday; estimate long-term order prediction on each Friday. (Open code: provide clear rules and procedures.)
Implement clan control	Director of Department 4: Now we emphasize the goal of the whole process, from acquiring orders to the whole supply chain. We have to unify our goals. Only when the unified goal is achieved can the value of each individual be realized. (Open code: instill a common goal.) Director of Strategy Department: Haier's organizational culture is the key. Although some of us might not understand the company's strategy fully, we executed quickly. Our colleagues from other departments are also willing to cooperate. (Open code: foster collective actions.)
The role of information processing capability	Director of Department 3: Technically, we implemented an information sharing platform. It helps to save money, save time and save efforts. Saving time means accessing to information is very fast. Saving money actually means information is shared and there is no necessary to purchase additional devices. In the past some tasks took us lots of efforts; for example, we had to negotiate a lot for the procurement task and it took us 40 days to deliver. But now it only takes one day. (Open code: improved operational efficiency.) Director of Infrastructure Operation and Maintenance Department: Once the sales order is fixed, support departments are ready to fulfill the sales order. For example, the information about the sales order is transferred to the systems of production department. A production plan will be generated accordingly. This information is simultaneously transferred to procurement department. The process is almost synchronized. (Open code: achievement of process synchronization.)

The company further developed clear rules and procedures, as well as implemented a performance management program to enhance the efficiency of information flow. Although the production task is complicated, establishing clear rules and procedures enables the complex task to be divided into smaller and simpler ones. As employees follow pre-specified rules and procedures, efficiency of the production process is largely improved. The company also developed the modularization of raw materials to reduce the complexity of the production process. Furthermore, with increasing process-related information recorded in the systems, the company implemented performance management program in production department to monitor and motivate employees. The performance of employees is evaluated based on certain evaluation criteria, and salary is influenced by the outcomes of the evaluation. Evaluation criteria, such as delivery on time and accuracy of order delivery, are pushing employees toward raising efficiency. As each employee becomes faster, the whole process speeds up.

Activities conducted in this task increased automation of information processing, leading to fluent information flows from the headquarters to manufacturing plants. As more production decisions are made automatically by the system without manual interventions, the company is able to enhance production efficiency. The relevant themes identified under 'Production Task' are summarized in Table 3.

A timeline of the order fulfillment process changes in Haier is provided in Appendix E. With above changes, the overall operation may be significantly improved. On one hand, the company is capable to detect market changes precisely. Salespersons are not only providing the accurate information of sales orders, but also offering their understanding of consumer preferences of the local market. As a result, the company has developed many innovative products such as the shrimp washer and the holiday freezer. On the other hand, the company is able to respond rapidly to market changes. The internal operational cycle has been largely reduced, allowing the company to deal with

Table 3 The relevant themes and data identified under 'production task'

Axial codes	Exemplary quote and open codes
Construct information processing network	Technical Leader 1 of Department 3: The systems are used to confirm and distribute production orders. After some manual adjustments, production orders will be transferred to factories every Wednesday night. Production department of each factory will organize its production accordingly. All of these are conducted by ERP systems. (Open code: leverage ERP system for production scheduling.) Director of Department 1: Our intranet covers all the factories. All information exchange is processed through information systems. (Open code: establish an information hub.)
Implement behavior control	Technical Leader of Department 4: The production task is driven by production plans. All the relevant systems will be used to monitor and execute tasks such as execution of production orders, management of material and product inventory, quality management and shipment. Usually we provide operation instructions to guide the normative operation. (Open code: provide clear rules and procedures.) Director of Department 2: We have developed a modularized production process. Now we attempt to increase the percentage of general modules. Some modules can be customized based on customer needs. As a result, we reduce the procurement and production cycle. (Open code: modularized production process.)
Implement outcome control	Director of Department 1: 'Individual-goal combination' is applied to all the staff, including staff from the supply chain and production department. There are specific performance management mechanisms for them. For manufacturing persons, their performance is evaluated based on the fulfillment of production goals. (Open code: implement comprehensive performance evaluation and provide incentives based on performance.) Technical Leader of Department 2: (The evaluation criteria for production department include,) first, deliver on time. Second, accuracy of order delivery. Furthermore, we also evaluate fulfillment of production capacity. For example, our production capacity is 5000 per day, but we only produce 4000 (which is undesirable). This can be considered as the usage ratio of production capacity. (Open code: define different performance evaluation criteria.)
The role of information processing capability	Director of Department 4: We emphasize the efficiency of the whole supply chain, from the input of customer orders to output of products. The cycle is only around 21 days. I cannot remember the exact number, but it does improve a lot. Our delivery cycle is the best in our (household appliance) industry. Our delivery efficiency is quite high. (Open code: improved operational efficiency.) Director of IT Infrastructure Management Department: We set up the '161 week order' mechanism in 2009. Some changes were made to our information systems accordingly. By doing this, we not only implemented zero inventory management, but achieved high-responsiveness toward customer demand. (Open code: improved operational efficiency.)

unpredictable market changes. For example, the company has successfully fulfilled an order placed by a German retailer within 2 days. Without an efficient order fulfillment process, it is less likely that the company will achieve this.

Discussion

To address the research question, we went through an iterative process between the relevant literature, the qualitative data and the emerging model. Subsequently a process model (see Figure 3) was developed. As the model suggests, operational agility is achieved through a two-step process – the construction of information processing network and the implementation of organizational control – to enhance the right information processing capability. Next, we explain the process model in detail, and show how the model enriches our understanding of operational agility.

Development of information sensitivity

Decision making for forecasting task was conducted within an environment filled with great uncertainty stemmed from environmental dynamism (Lee & Grover, 1999). One specific type of environmental dynamism is termed demand uncertainty, which indicates the changes in demand for products and the inability to accurately predict these fluctuations (Premkumar *et al*, 2003). It captures one of the natures of the contemporary business landscape, where customer needs keep changing rapidly (Overby *et al*, 2006). The metaphor of 'flying targets' from Haier's CEO attests to this challenge facing the company.

To cope with the information needs emanating from environmental dynamism, the *information star* was established and the *outcome and self controls* were implemented. As the information processing network is characterized by high level of centralization and low level of hierarchy, we labeled it as 'information star'. This kind of network is built by establishing direct information exchange channels to

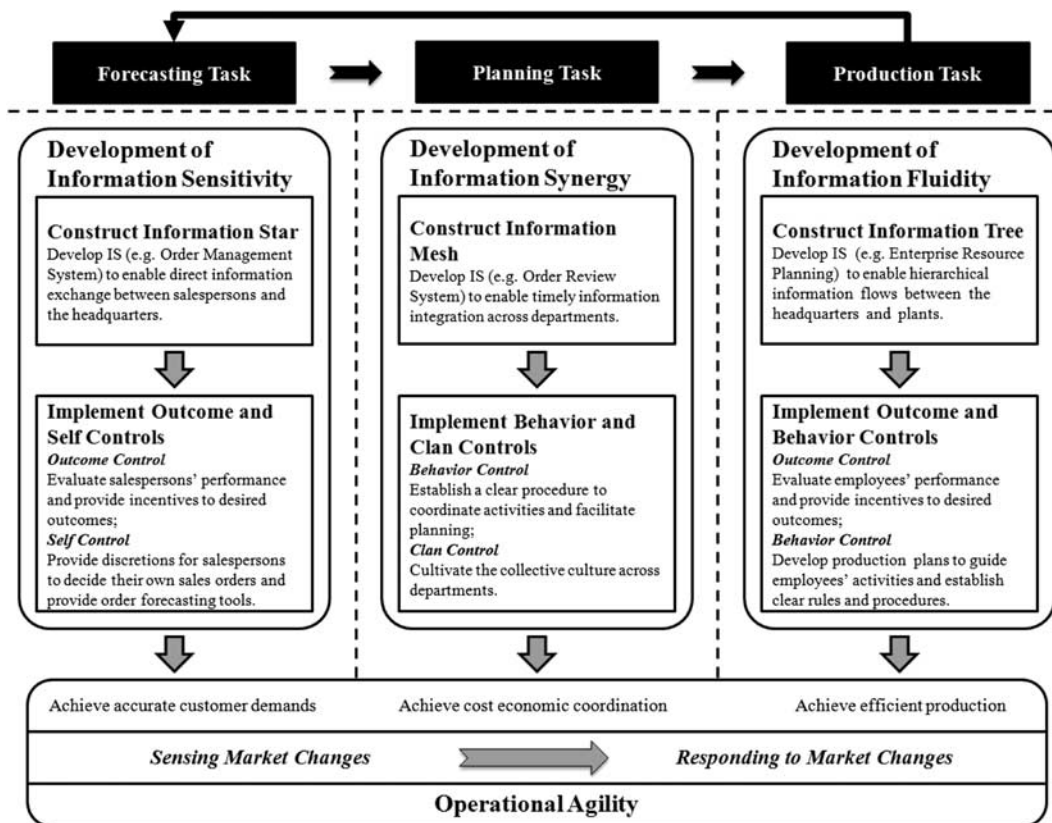


Figure 3 A process model of developing information processing capability for operational agility.

connect salespersons in point of sales and sales department in the headquarters. This structure is similar in spirit to the hybrid communication network, in that it allows both efficiency and flexibility simultaneously (Brown & Eisenhardt, 1998). The centralized structure is efficient when information interactions occur vertically (e.g. information sent from central nodes to peripheral nodes) (Ahuja & Carley, 1999); low level of hierarchy representing a flat structure is flexible in facilitating mutual adjustment between the center and peripheral nodes. Examples of enterprise information systems that enable the construction of information star include order management systems and intranet portals. To ensure the optimal performance of salespersons at the point of sale, the outcome and self controls were implemented. By developing the performance management program to evaluate salespersons' performance and providing incentives to desired performance, outcome control was executed (Eisenhardt, 1985). Meanwhile, the company provided sufficient discretions for salespersons to make sales decisions, and offered tools (such as the order forecasting tools) to support their tasks, which indicates the execution of self control (Manz *et al*, 1987). These control mechanisms not only ensure that salespersons behave in a way that is consistent with organizational objectives, but also provide them with ample room to react to market changes.

Accordingly, the ability to detect changes of customer needs readily by collecting and analyzing relevant information was developed, and named *information sensitivity*. This was developed by constructing the information star and implementing the outcome and self controls. The information sensitivity is similar to the 'market sensitivity' (Christopher, 2000) in supply chain research, which is considered as a key determinant of agile supply chain. This capability helps an organization to make accurate sales decisions. It further contributes to the sensing component of operational agility by enhancing the market intelligence capability (Overby *et al*, 2006).

Development of information synergy

The great uncertainty that confronted the decision making for planning task originated mainly from task interdependence (Gattiker, 2007; Sharma & Yetton, 2007), since this task required tight collaboration across departments. Actions taken within one department affect the actions and work outcomes of other departments (McCann & Ferry, 1979) in this task. This interdependence among departments increases the likelihood that changing conditions in one department require other departments to adjust (Thompson, 1967; Sharma *et al*, 2008). To achieve this type of adjustment, each department should be aware

of information about other departments during decision making (Gattiker, 2007).

The requirement to deal with task interdependence drove the establishment of *information mesh* and the implementation of *behavior and clan controls*. The information processing network characterized by a low level of centralization and hierarchy is named 'information mesh'. The information mesh is built by establishing highly connected information exchange channels across diverse departments, to enable timely information integration. This structure is similar to the 'small world' network (Kwon *et al*, 2007) or the organic communication network (Ahuja & Carley, 1999). It supports routine communications among actors or units, and enhances mutual dependence among interconnected nodes, especially when tight collaboration is desired to connect value chains within the organization (Newman, 2004). Electronic calendars, workflow systems and knowledge management systems are examples of enterprise information systems that enable the construction of information mesh. To ensure decision making is based on consistent information, behavior and clan controls were implemented. The behavior control was used through establishing a clear procedure to coordinate activities across departments and conducting regular meetings to facilitate planning (Eisenhardt, 1985). The clan control was executed by propagating the 'one common goal' belief and cultivating the collective culture among departments (Ouchi, 1979; Ouchi, 1980). These control mechanisms were implemented to optimize the flexibility of collective actions, while maintaining efficient cooperation among different departments during decision making.

Consequently, the ability to integrate information for effective communication and collaboration across unit boundaries was developed, and named the *information synergy*. According to Dewett and Jones (2001), information synergy can be achieved when IT allows the different units to adjust their actions to the needs of the other units on an ongoing basis. Our findings show that information synergy was developed through constructing the information mesh and implementing the behavior and clan controls. Given the crucial role played by IT in this task, this finding is consistent with Dewett and Jones (2001). This capability helps organization achieve cost economic coordination across diverse departments. It further contributes to the sensing and responding components (Overby *et al*, 2006) of operational agility, in that it allows the organization to achieve a holistic understanding of environments and create a coordinated responding plan to guide organizational actions.

Development of information fluidity

Decision making for production task was confronted with great uncertainty, mainly stemming from task complexity (Lee & Grover, 1999). It is found that task becomes more complex as the number of factors and their interactions increase (Ramamurthy, 1990). Task complexity captures the characteristic of the production process

through which various types and a large number of products are produced. This complex product structure constitutes one of the key determinants of production complexity (Calinescu *et al*, 1998). As product structure becomes complex, so distinct information cues must be processed in decision making.

The *information tree* was established, and the *outcome and behavior controls* were implemented to fulfill information needs. The information processing network characterized by high level of centralization and hierarchy is named 'information tree'. The information tree is built by establishing hierarchical information disseminating channels between the headquarters and manufacturing plants. This is similar to the 'moderate scale free' network (Kwon *et al*, 2007) or the mechanistic communication network (Tushman, 1979). Among systems of a given size and complexity, it requires less information exchange among actors (Simon, 1977) than do the information mesh, so that less information cues are generated, reducing the need of manual interventions. The dispersed manufacturing plants of the case company rely on this communication hierarchies for efficiency in communication and coordination (Ahuja & Carley, 1999). ERP systems and supply chain management systems are examples of enterprise information systems that enable the construction of information tree. Accordingly, the behavior and outcome controls were implemented to coordinate among members during the production process. Behavior control was executed by developing production plans to guide individuals' activities, utilizing a modularization method and establishing clear rules and procedures (Kirsch, 1997). To implement outcome control, employees' performance was evaluated, and incentives were provided to a desired outcome (Eisenhardt, 1985). These control mechanisms allow the company to pre-specify desired behaviors, as well as outcomes, which reduces the demand for decision making and enhances operational efficiency.

The information processing capability developed in this task is named *information fluidity*, which indicates the ability to facilitate efficient flowing of information by increasing automation in processing information. This is achieved through constructing the information tree and implementing the outcome and behavior controls. Existing research finds that information systems (e.g. ERP) enhance task efficiency by improving information flow (Gattiker & Goodhue, 2005). This is corroborated by our study that information systems allow organization to achieve information fluidity for improved efficiency. The information fluidity contributes to the responding component of operational agility by enhancing the production capability (Overby *et al*, 2006).

Operational agility may be defined as the ability to achieve accurate customer demands, cost economic coordination and efficient production in the context of order fulfillment (e.g. Flidner, 2003). By developing information sensitivity, information synergy and information fluidity, firms may achieve accurate market demands, cost economic coordination and efficient production in order fulfillment. Since the

Table 4 A five-step practical guide*Developing information processing capability for operational agility*

Step 1: Investigate the information needs of business process. The first step requires the focal organization to understand the information needs of business process. Information needs stem from task environment, task interdependence and task characteristics. Is the task environment dynamic? Is it a task that requires tight cooperation among multiple stakeholders? Is the task complex? The focal organization must be able to address these questions before it is ready to develop information processing capability. The outcome of this step allows the organization to concentrate on fewer aspects of business process, but the most salient ones.

Step 2: Evaluate the cost and benefit. The second step requires the organization to evaluate the cost and benefit of developing information processing capability. While the information synergy is capable of dealing effectively with greater amounts of uncertainty than information fluidity, there are costs associated with this increased information processing capacity. It takes more time, effort and energy to manage the information mesh and implement the behavior and clan controls. Thus, the benefits of increased information processing capability must be weighed against the associated costs of managing the information processing network and implementing control mechanisms. The outcome of this step makes the organization narrow down the feasible alternatives.

Step 3: Construct an information processing network. The third step requires the organization to construct an appropriate information processing network. It involves evaluating the current information processing network and generating a new information processing network. The deliverables of steps 1 and 2 will serve as the input for this step. Accordingly, the focal organization must evaluate whether the current information processing network (i.e. existing information and communication systems) is capable of fulfilling the information needs estimated in step 1. If the existing one fails, then more effort should be devoted to adjusting the existing one or creating a new network. It might be feasible to make some changes to the existing information processing network, such as integrating the legacy systems. It might also be workable to tailor a new information processing network for the business process. The focal organization should evaluate the alternative approaches to decide on the most suitable one.

Step 4: Implement appropriate control mechanisms. This step requires the organization to implement appropriate control mechanisms to influence employee's information processing behavior. The focal organization should first choose between formal controls and informal controls. Formal controls require a prescription of expected behaviors or outcomes. The focal organization can choose behavior control or outcome control to implement formal controls. Informal controls are based on social or people strategies, which provide the organization with flexibility in dealing with uncertainty. Accordingly, clan control and self control may be used to implement informal controls.

Step 5: Align control with information processing network. Finally, the organization should adjust control mechanisms to fit the information processing network. Since the information processing network differs in terms of the flexibility in dealing with uncertainty, so do control mechanisms. Informed by this study, different information processing networks work better with specific control mechanisms. For example, the information mesh is better with behavior and clan controls, while the information tree works better with outcome and behavior controls. It is this very combination that allows the company to achieve operational agility. Therefore, the company should adjust the control mechanisms implemented to make it suitable for the information processing network.

three information processing capabilities developed are iteratively applied in these tasks (see the black arrows shown in Figure 3), they are routinized for repeated application. Thus, our model suggests that operational agility can be attained through developing three routinized information processing capabilities. On the basis of these findings, we provide a practical guide for developing the information processing capability for operational agility (see Table 4), which helps practitioners improve business operations.

Conclusion

Theoretical contributions

This study provides potential contributions to the literature in several ways. First, it contributes to the agility literature by developing a process model to show how operational agility is achieved. Past research on operational agility mainly centers on identifying and validating the enablers or antecedents of operational agility (e.g. Raschke, 2007; Tallon, 2008). However, the process

through which operational agility is developed has received little attention. By adopting case research methods, this study empirically examines the process of operational agility development. A process model is developed and it shows that operational agility is achieved through a two-step process that enhances the right information processing capability. We hope that the findings of this study can address the aforementioned theoretical gap.

Second, this research also contributes to the information processing view of firms. It makes a contribution by depicting a two-step process – the construction of information processing network and the implementation of organizational control – to show how information processing capability is developed. In this way, some new concepts such as the three types of information processing capabilities are identified. It is hoped that these concepts can form a foundation for future research on the information processing view. Furthermore, it corroborates previous findings, namely that information processing capability plays a foundational role in developing a firm's capabilities to improve firm performance (Mithas *et al*, 2011).

Finally, previous agility research focuses on investigating the role of IT assets (Mithas *et al*, 2011), while the current study offers an attempt to study agility from an information processing perspective. Agility is decomposed into two components: sensing and responding (Overby *et al*, 2006). The sensing components rely on organizational capability in processing information from internal and external environments, so that accurate understanding of environments can be attained before making decisions. The responding components depend on the ability to transform information into an actionable plan and disseminate information to coordinate activities. Thus, the information processing perspective provides a viable theoretical lens to study agility. We hope that this study can provide a sound example of combining the information processing perspective and agility literature and stimulate more efforts to achieve this end.

Limitation and future research

The current study has two main limitations. First, by investigating the process of developing information processing capability for operational agility, some new concepts are developed, such as information sensitivity, information synergy and information fluidity. Although the preliminary definitions and descriptions of these concepts are provided, their measurements are missing in this study. Future research may attempt to operationalize the new concepts and collect survey data to empirically validate the findings from this study. Second, the findings of this study may be affected by the retrospective nature of the interview data. A disadvantage of retrospective responses is that they may suffer from errors of recall (Glick *et al*, 1990). We have sought to minimize this effects in two ways: on the one hand, we chose informants who were personally involved in order fulfillment process changes (e.g. Pan *et al*, 2007); on the other hand, a systematic data verification procedure was followed to ensure data triangulation (Klein & Myers, 1999).

About the Authors

Pei-Ying Huang is a Ph.D. candidate of the Department of Information Systems in School of Computing at National University of Singapore. His research interests include enterprise agility, organizational information processing, and other IT-related management issues. He has published his works in International Conference on Information Systems.

Shan-Ling Pan (Shan L. Pan) is Professor of Information Systems and Technology Management at the Australian School of Business, University of New South Wales, Australia. His research work has been published in various

The current study enhances our understanding of the impact of information processing on operational agility. According to the agility literature, this construct consists of three interrelated capabilities: customer agility, operational agility, and partnering agility (Sambamurthy *et al*, 2003). While we are only concerned with operational agility in the current study, research on the impact of information processing on customer and partnering agility is also needed. Customer agility is about co-opting customers and partnering agility is about leveraging assets, knowledge, and competencies of partners. From the information processing perspective, customer agility is more likely to be achieved by processing customer information to support decision making. The ability to integrate partners' information to support decision making is more likely to contribute to partnering agility. Thus, more studies are needed to contribute to the agility literature.

Given that today's business environments have become increasingly turbulent, many IS researchers are motivated to study how IT enables contemporary organizations to adapt to environmental changes (e.g. Sambamurthy *et al*, 2003; Lu & Ramamurthy, 2011). Most previous research focuses on investigating the role of IT assets (e.g. IT investments). In contrast, this study draws attention to the role of information processing in enabling organizational adaptation. Kohli and Grover (2008) point out that the way in which information processing capabilities may enhance digital business capabilities is an increasingly important topic, which has thus far been neglected. This study responds to their call. Nonetheless, more research on the role of information processing in contemporary organizations is needed.

Acknowledgements

Funding for this research was provided by National Natural Science Foundation of China (NSFC 71228201, 71072021, 71172176).

top tier journals such as Information Systems Research, Journal of AIS, European Journal of Information Systems, Information Systems Journal, etc.

Tao Hua Ouyang is a professor of School of Economics and Management at Beihang University, China. She received master degree of Economics from Wuhan University, China and Ph.D. degree of Management Science from Kobe University of Japan. Her research interests include strategic management and modern business management.

References

- AHUJA M and CARLEY K (1999) Network structure in virtual organizations. *Organization Science* **10**(6), 741–757.
- BAKER WE (1992) The network organization in theory and practice. In *Networks and Organizations: Structure, Form and Action* (NOHRIA N and ECCLES R, Eds), Harvard Business School Press, Boston, MA.
- BROWN SL and EISENHARDT KM (1998) *Competing on the Edge: Strategy as Structured Chaos*. Harvard Business School Press, Boston, MA.

- CALINESCU A, EFSTATHIOU J, SCHIRN J and BERMEJO J (1998) Applying and assessing two methods for measuring complexity in manufacturing. *The Journal of the Operational Research Society* **49**(7), 723–733.
- CARDINAL LB, SITKIN SB and LONG CP (2004) Balancing and rebalancing in the creation and evolution of organizational control. *Organization Science* **15**(4), 411–431.
- CHRISTOPHER M (2000) The agile supply chain: competing in volatile markets. *Industrial Marketing Management* **29**(1), 37–44.
- CROXTON KL (2003) The order fulfillment process. *International Journal of Logistics Management, The* **14**(1), 19–32.
- DALNE H (2011) Operational agility: From supply chain to integrated value chain. In *Growing Beyond: How High Performers are Competing for Growth in Difficult Times*. Ernst&Young, London.
- DEWETT T and JONES GR (2001) The role of information technology in the organization: a review, model, and assessment. *Journal of Management* **27**(3), 313–346.
- DOVE R (2001) *Response Ability: The Language, Structure, and Culture of The Agile Enterprise*. John Wiley & Sons, New York.
- EISENHARDT KM (1985) Control: organizational and economic approaches. *Management Science* **31**(2), 134–149.
- EISENHARDT KM (1989) Building theories from case study research. *Academy of Management Review* **14**(4), 532–550.
- FERDOWS K, LEWIS MA and MACHUCA JA (2004) Rapid-fire fulfillment. *Harvard Business Review* **82**(11), 104–117.
- FLIEDNER G (2003) CPFR: an emerging supply chain tool. *Industrial Management & Data Systems* **103**(1), 14–21.
- FOGARTY DW, BLACKSTONE JH and HOFFMANN TR (1991) *Production & Inventory Management*. South-Western Publishing Company, Nashville, TN.
- FONTANA A and FREY JH (2000) The interview: from structured questions to negotiated text. In *Handbook of Qualitative Research* (DENZIN NK and LINCOLN YS, Eds) 2nd edn, pp 645–672, Sage, Thousand Oaks, CA.
- FREEMAN LC (1979) Centrality in social networks conceptual clarification. *Social Networks* **1**(3), 215–239.
- FULK J and DeSANCTIS G (Eds) (1999) Articulation of communication technology and organizational form. In *Shaping Organization form: Communication, Connection, and Community*, pp 5–32, Sage, Thousand Oaks, CA.
- GALBRAITH JR (1973) *Designing Complex Organizations*. Addison-Wesley Longman Publishing Co., Inc, Boston, MA.
- GALBRAITH JR (1974) Organization design: an information processing view. *Interfaces* **4**(3), 28–36.
- GALLIERS RD (2007) Strategizing for agility: confronting information systems inflexibility in dynamic environments. In *Agile Information Systems* (DESOUZA K, Ed), pp 1–15, Butterworth-Heinemann, Elsevier Inc, Burlington, MA.
- GATTIKER T (2007) Enterprise resource planning (ERP) systems and the manufacturing-marketing interface: an information-processing theory view. *International Journal of Production Research* **45**(13), 2895–2918.
- GATTIKER TF and GOODHUE DL (2005) What happens after ERP implementation: understanding the impact of interdependence and differentiation on plant-level outcomes. *MIS Quarterly* **29**(3), 559–585.
- GLENN M (2009) Organisational agility: how business can survive and thrive in turbulent times. A Report from the Economist Intelligence Unit, *The Economist*, March.
- GLICK WH, HUBER GP, MILLER CC, DOTY DH and SUTCLIFFE KM (1990) Studying changes in organizational design and effectiveness: retrospective event histories and periodic assessments. *Organization Science* **1**(3), 293–312.
- HUMMON NP and FARARO TJ (1995) Actors and networks as objects. *Social Networks* **17**(1), 1–26.
- JARVENPAA SL, KNOLL K and LEIDNER DE (1998) Is anybody out there?: antecedents of trust in global virtual teams. *Journal of Management Information Systems* **14**(4), 29–64.
- KIRSCH LJ, KO D-G and HANEY MH (2010) Investigating the antecedents of team-based clan control: adding social capital as a predictor. *Organization Science* **21**(2), 469–489.
- KIRSCH LJ, SAMBAMURTHY V, KO D-G and PURVIS RL (2002) Controlling information systems development projects: the view from the client. *Management Science* **48**(4), 484–498.
- KIRSCH LS (1997) Portfolios of control modes and IS project management. *Information Systems Research* **8**(3), 215–239.
- KLEIN HK and MYERS MD (1999) A set of principles for conducting and evaluating interpretive field studies in information systems. *MIS Quarterly* **23**(1), 67–93.
- KODAMA M (2005) How two Japanese high-tech companies achieved rapid innovation via strategic community networks. *Strategy & Leadership* **33**(6), 39–47.
- KOHLI R and GROVER V (2008) Business value of IT: an essay on expanding research directions to keep up with the times. *Journal of the Association for Information Systems* **9**(1), 23–39.
- KRAUT R, STEINFELD C, CHAN A, BUTLER B and HOAG A (1999) Coordination and virtualization: the role of electronic networks and personal relationships. *Organization Science* **10**(6), 722–740.
- KWON D, OH W and JEON S (2007) Broken ties: the impact of organizational restructuring on the stability of information-processing networks. *Journal of Management Information Systems* **24**(1), 201–231.
- LEE CC and GROVER V (1999) Exploring mediation between environmental and structural attributes: the penetration of communication technologies in manufacturing organizations. *Journal of Management Information Systems* **16**(3), 187–217.
- LEIFER R and MILLS PK (1996) An information processing approach for deciding upon control strategies and reducing control loss in emerging organizations. *Journal of Management* **22**(1), 113.
- LIM ETK, TAN CW, CYR D, PAN SL and XIAO B (2012) Advancing public trust relationships in electronic government: the Singapore E-filing journey. *Information Systems Research* **23**(4), 1110–1130.
- LU Y and RAMAMURTHY K (2011) Understanding the link between information technology capability and organizational agility: an empirical examination. *MIS Quarterly* **35**(4), 931–954.
- MANI D, BARUA A and WHINSTON A (2010) An empirical analysis of the impact of information capabilities design on business process outsourcing performance. *MIS Quarterly* **34**(1), 39–62.
- MANZ CC, MOSSHOLDER KW and LUTHANS F (1987) An integrated perspective of self-control in organizations. *Administration & Society* **19**(1), 3–24.
- MCCANN JE and FERRY DL (1979) An approach for assessing and managing inter-unit interdependence. *The Academy of Management Review* **4**(1), 113–119.
- MITHAS S, RAMASUBBU N and SAMBAMURTHY V (2011) How information management capability influences firm performance. *MIS Quarterly* **35**(1), 237–256.
- NEWMAN MEJ (2004) Fast algorithm for detecting community structure in networks. *Physical Review E* **69**(6), 066133.
- OUCHI WG (1979) A conceptual framework for the design of organizational control mechanisms. *Management Science* **25**(9), 833–848.
- OUCHI WG (1980) Markets, bureaucracies, and clans. *Administrative Science Quarterly* **25**(1), 129–141.
- OVERBY E, BHARADWAJ A and SAMBAMURTHY V (2006) Enterprise agility and the enabling role of information technology. *European Journal of Information Systems* **15**(2), 120–131.
- OWEN-SMITH J and POWELL WW (2004) Knowledge networks as channels and conduits: the effects of spillovers in the Boston biotechnology community. *Organization Science* **15**(1), 5–21.
- PAN SL, PAN G, CHEN AJW and HSIEH MH (2007) The dynamics of implementing and managing modularity of organizational routines during capability development: insights from a process model. *IEEE Transactions on Engineering Management* **54**(4), 800–813.
- PAN SL, PAN G and LEIDNER D (2012) Crisis response information networks. *Journal of the Association for Information Systems* **13**(1), 518–555.
- PAN SL, PAN G, NEWMAN M and FLYNN D (2006) Escalation and de-escalation of commitment to information systems projects: insights from a project evaluation model. *European Journal of Operational Research* **173**(3), 1139–1160.
- PAN SL and TAN B (2011) Demystifying case research: a structured-pragmatic-situational (SPS) approach to conducting case studies. *Information and Organization* **21**(3), 161–176.
- PENTLAND BT (1999) Building process theory with narrative: from description to explanation. *Academy of Management Review* **24**(4), 711–724.
- PORTER ME (2001) Strategy and the internet. *Harvard Business Review* **79**(3), 62–79.
- PREMKUMAR G, RAMAMURTHY K and SAUNDERS CS (2003) Information processing view of organizations: an exploratory examination of fit in the context of interorganizational relationships. *Journal of Management Information Systems* **22**(1), 257–294.

- RAMAMURTHY K (1990) *Role of Environmental, Organizational and Technological Factors in Information Technology Implementation in Advanced Manufacturing: An Innovation Adoption-diffusion Perspective*. University of Pittsburgh, Pittsburgh, PA.
- RASCHKE RL (2007) An empirical analysis of business process agility: Examining the relationship of IT on business process agility and the effects of business process agility on process outcomes. Arizona State University.
- RASCHKE RL and DAVID JS (2005) Business process agility. In *Americas Conference on Information Systems, 2005 Proceedings*, Omaha, Nebraska.
- RUSTAGI S, KING WR and KIRSCH LJ (2008) Predictors of formal control usage in IT outsourcing partnerships. *Information Systems Research* **19**(2), 126–143.
- SAMBAMURTHY V, BHARADWAJ A and GROVER V (2003) Shaping agility through digital options: reconceptualizing the role of information technology in contemporary firms. *MIS Quarterly* **27**(2), 237–263.
- SHARMA R and YETTON P (2007) The contingent effects of training, technical complexity, and task interdependence on successful information systems implementation. *MIS Quarterly* **31**(2), 219–238.
- SHARMA R, YETTON PW and ZMUD RW (2008) Implementation costs of IS-enabled organizational change. *Information and Organization* **18**(2), 73–100.
- SIMON HA (1977) *The New Science of Management Decision*. Prentice Hall, New Jersey.
- STRAUSS A and CORBIN JM (1990) *Basics of Qualitative Research: Grounded Theory Procedures and Techniques*. Sage Publications, Inc, Thousand Oaks, CA.
- TALLON PP (2008) Inside the adaptive enterprise: an information technology capabilities perspective on business process agility. *Information Technology and Management* **9**(1), 21–36.
- THOMPSON JD (1967) *Organizations in Action: Social Science Bases of Administrative Theory*. McGraw-Hill, New York.
- TURNER KL and MAKHJIA MV (2006) The role of organizational controls in managing knowledge. *Academy of Management Review* **31**(1), 197–217.
- TUSHMAN ML (1979) Work characteristics and subunit communication structure: a contingency analysis. *Administrative Science Quarterly* **24**(1), 82–98.
- TUSHMAN ML and NADLER DA (1978) Information processing as an integrating concept in organizational design. *The Academy of Management Review* **3**(3), 613–624.
- VAN OOSTERHOUT M, WAAARTS E and VAN HILLEGERBERG J (2006) Change factors requiring agility and implications for IT. *European Journal of Information Systems* **15**(2), 132–145.
- WALSHAM G (1995) Interpretive case studies in IS research: nature and method. *European Journal of Information Systems* **4**(2), 74–81.
- WALSHAM G (2006) Doing interpretive research. *European Journal of Information Systems* **15**(3), 320–330.
- WANG ET, TAI JC and GROVER V (2013) Examining the relational benefits of improved interfirm information processing capability in buyer – supplier dyads. *Management Information Systems Quarterly* **37**(1), 149–173.
- WEILL P, SUBRAMANI M and BROADBENT M (2002) Building IT infrastructure of strategic agility. *Sloan Management Review* **44**(1), 57–65.
- WIESENFELD BM, RAGHURAM S and GARUD R (1999) Communication patterns as determinants of organizational identification in a virtual organization. *Organization Science* **10**(6), 777–790.
- WU J and LI R (2008) Modeling the relationship between IT capability, uncertainty and organizational performance: an empirical study. In *Advanced Management of Information for Globalized Enterprises, AMIGE 2008* (ZHAO JL, CHEN J, HOFMANN P, ZHANG D, HUANG J and HE Z, Eds), pp 1–5, IEEE, Tianjin, China.
- YIN R (2003) *Applications of Case Study Research*. Sage Publications, Inc.
- YONG H (2008) *Haier Reaches Higher*. Zhejiang people's publishing house, Zhejiang, China.

Appendix A

A list of general interview questions

- How does the internal operation facilitate the company sense and respond to market changes?
- What are the tasks and procedures of the internal operation? Who is involved? How does it proceed?
- What are the challenges of the internal operation? Why? How may these challenges be overcome?
- What is the role of IT in the internal operation? Why?
- What are the information systems involved in the internal operation? What are they used for?
- What are the information needs of the internal operation? Why? How are these needs fulfilled?
- How can it be ensured that the internal operation is functioning well?
- Which task do you attend? What is your job?
- Do you think the information systems fulfill the information needs of your daily job? How do they achieve this?

Appendix B

Table B1 Breakdown of group/individual interviews

Group identifier	Number of interviewees	Level of interviewees			Position of interviewees	Relevance (%) ^a
		Top management	Middle management	Senior staff		
1	3	0	3	0	Director of Strategy Department; Director of Department 1; Director of Infrastructure Operation and Maintenance Department	66.0
2	2	0	2	0	Director of Infrastructure Operation and Maintenance Department; Director of IT Infrastructure Management Department	63.2
3	3	0	2	1	Director of Department 1; Technical Leader of Department 1; Director of Strategy Department	84.2
4	2	0	2	0	Director of Strategy Department; Director of Department 1	96.4
5	2	0	1	1	Director of Department 2; Technical Leader of Department 2	82.0
6	4	0	3	1	Director of Department 4; Project Manager of Department 4; Technical Leader of Department 4; Director of Strategy Department	78.6
7	3	0	2	1	Director of Department 3; Technical Leader 1 of Department 3; Director of Strategy Department	57.1
8	3	0	1	2	Director of Department 3; Technical Leader 1 of Department 3; Technical Leader 2 of Department 3	62.5
9	4	1	3	0	Chief Information Officer (CIO); Director of Strategy Department; Director of Department 1; Director of Department 3	59.1
10 ^b	1	0	1	0	Project Manager of Department 4;	95.1
11 ^b	1	0	1	0	Project Manager of Department 4;	94.4
12	2	1	1	0	Director of Haier University; Director of Supply Chain Department	85.2
Person-Time	30	2	22	6	N/A	N/A

^aRelevance represents the extent to which the content of a specific interview is relevant to the themes studied. It is calculated by dividing the number of individual statements relevant to the themes studied by the total number of individual statements made (see Appendix C).

^bGroups 10 and 11 are individual interviews.

Appendix C

Table C1 Summary of axial codes

<i>Context</i>	<i>Axial codes</i>	<i>Hits^a(%)</i>	<i>Intensity^b</i>
<i>Decision making for forecasting</i>	Construct information processing network (information star)	83.3	34
	Implement outcome control	75.0	71
	Implement self control	75.0	41
	The role of information processing capability (information sensitivity)	75.0	29
<i>Decision making for planning</i>	Construct information processing network (information mesh)	58.3	32
	Implement behavior control	58.3	34
	Implement clan control	58.3	26
	The role of information processing capability (information synergy)	50.0	25
<i>Decision making for production</i>	Construct information processing network (information tree)	50.0	20
	Implement behavior control	58.3	37
	Implement clan control	75.0	26
	The role of information processing capability (information sensitivity)	50.0	24

^aHits represent the number of interviews (group interviews or individual interviews) that have mentioned about a particular theme (or axial code) out of the total 12 interviews.

^bIntensity represents the total number of individual statements of all interviews that relate to a particular theme (or axial code).

Appendix D

Table D1 A sample of 'statistics for group-6 interview'

<i>Open codes</i>	<i>Axial codes</i>	<i>Intensity^a</i>	<i>Subtotal</i>
<i>Forecasting task</i>			
The needs of understanding the market accurately	Information processing requirements of forecasting	4	7
Complex sales network		3	
Implement system integration with downstream distributors	Construct information processing network (information star)	5	8
Establish multiple communication channels between salespersons and the headquarters		3	
Implement comprehensive performance evaluation and provide incentives based on performance	Implement outcome control	4	4
Individual empowerment	Implement self control	3	3
Achievement of accurate prediction	The role of information processing capability (information sensitivity)	2	2
<i>Planning task</i>			
N/A	Information processing requirements of planning	0	0
Implement internal systems integration	Construct information processing network (information mesh)	3	3
Regular meetings	Implement behavior control	3	8
Implement internal systems integration		5	
Instill a common goal	Implement clan control	7	8
Forster collective actions		1	
N/A	The role of information processing capability (information synergy)	0	0
<i>Production task</i>			
Complex product structure	Information processing requirements of production	2	3

Table D1: (Continued)

Open codes	Axial codes	Intensity ^a	Subtotal
The need for efficient production		1	
Establish an information hub	Construct information processing network (information tree)	2	2
Implement comprehensive performance evaluation and provide incentives based on performance	Implement outcome control	2	2
Strict schedule control	Implement behavior control	7	9
Modularized production process		2	
Improved operational efficiency	The role of information processing capability (information fluidity)	3	3
Demonstration of operational agility	N/A	4	4
Subtotal			66
Top management decision making	Irrelevant themes	2	18
Relationship with distributors		4	
Relationship with suppliers		1	
Business process re-engineering		10	
IT project management		1	
Total			84
Relevance ^b			78.6%

^aIntensity represents the total number of individual statements made that relate to a particular theme (or axial code).

^bRelevance represents the extent to which the content of a specific interview is relevant to the themes studied. It is calculated by dividing the number of individual statements relevant to the themes studied by the total number of individual statements made.

Appendix E

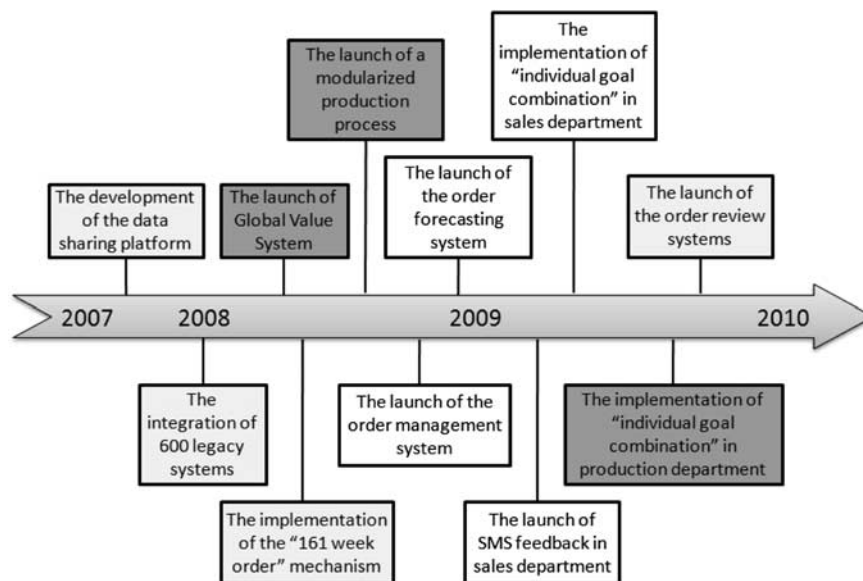


Figure E1 A timeline of order fulfillment process changes in Haier.

Notes: The two-step process may not be obvious as we mixed the actions of forecasting, planning, and production task in a timeline. We use different colors of the box to differentiate the actions.

A white color box refers to an action that enhances information processing capability of the forecasting task.

A light gray color box refers to an action that enhances information processing capability of the planning task.

A dark gray color box refers to an action that enhances information processing capability of the production task.

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.